Total no. of Pages: ..01.....

Roll no. ....

BTECH **Semester: ...III** **- Course work** **ECAM**

END-SEMESTER THEORY EXAMINATION, NOV-DEC,2024

Course Code: EAEPC302/ EAEPC05
time: 03 Hours

Course Title SIGNALS AND SYSTEMS
Max Marks: 50

Note: Attempt all the five questions. Missing data/ information (if any), maybe suitably assumed & mentioned in the answer.

Q 1	**Attempt any 2 parts of the following**		
1a	Explain the following functions- (i)unit step (ii)unit ramp (iii)impulse function	5	CO1
1b	explain continuous time and discrete time functions.give one example.	5	CO1
1c	find solution to the following (i) $\sum_{n=-\infty}^{+\infty} \delta[n-2]\text{Sin } 2n$ (ii) $\sum_{n=0}^{+\infty} \delta[n]exp\,(2n)$	5	CO1
Q 2	**Attempt any 2 parts of the following**		
2a	Determine the values of power and energy of the signal $x[n]=(1/3)^{n}u[n]$	5	CO2
2b	(i)explain properties of an LTI system (ii)determine if $y[n=x[n]+\frac{1}{x[n-1]}$ and $y[n]=x^{2}[n]$ are linear or nonlinear functions.	5	CO2
2c	Determine the convolution of following sequence $x[n]=h[n]=\{1,2,-1\}$	5	CO2
Q 3	**Attempt any 2 parts of the following**		
3a	State Dirichlet's conditions for a function to be expanded as a Fourier series	5	CO3
3b	State fourier integral and inverse fourier integral theoram	5	CO3
3c	Prove the linearity and scaling property of FT.	5	CO3
Q 4	**Attempt any 2 parts of the following**		
4a	Find Laplace transform of cos(at)u(t)	5	CO4
4b	Explain Laplace transform and its time shift and convolution properties.	5	CO4
4c	Find out the transfer function of a low pass RC circuit using laplace transform.	5	CO4
Q 5	**Attempt any 2 parts of the following**		
5a	A causal LTI system is described by the described by the difference equation $y[n]=y[n-1]+y[n-2]+x[n-1]$ where x[n] is the input and y[n] is the output. Find (i) The system function $H(z)=Y(z)/X(z)$ for the system. Plot the poles and zeroes of the H(z) and indicate ROC.State the stability of the H(z).	5	CO5
5b	Write definition of Z transform and derive the equation for inverse Z transform.	5	CO5
5c	Find Z transform and ROC of the causal sequence $x[n]=\{1,0,3,-1,2\}$	5	CO5